CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $5x^2 + 6x + 1 = 0$, the value of b is: 1

(a) -6 **(b)** 5 **(c)** 1 **(d)** 6

2. The roots of $(x + 2)(x - 6) = 0$ are: 1

(a) 2 and -6 **(b)** -2 and -6 **(c)** 4 and -12 **(d)** -2 and 6

3. The discriminant of $3x^2 - 2x - 1 = 0$ is: 1

(a) -16 **(b)** 16 **(c)** -8 **(d)** 10

4. The nature of the roots of $2x^2 - 4x + 2 = 0$ is: 1

(a) three real roots **(b)** two equal real roots
(c) no real roots **(d)** two distinct real roots

5. Which of the following is a quadratic equation? 1

(a) $x + 2 = 9$ **(b)** $x^3 + 2x + 9 = 0$ **(c)** $x^2 + 2x + 9 = 0$ **(d)** $2x + 9 = 0$

6. Which of the following is a root of $x^2 - 4x + 3 = 0$? 1

(a) 1 **(b)** 0 **(c)** 2 **(d)** -1

7. In $4x^2 - 8x + 3 = 0$, the value of b is: 1

(a) 3 **(b)** -8 **(c)** 8 **(d)** 4

8. The roots of $(x - 1)(x - 7) = 0$ are: 1

(a) 1 and -7 **(b)** -1 and -7 **(c)** 8 and 7 **(d)** 1 and 7

9. The discriminant of $x^2 + x + 3 = 0$ is: 1

(a) -11 **(b)** 11 **(c)** -5 **(d)** 13

10. Which of the following is a quadratic equation? 1

(a) $x - 1 = 3$ **(b)** $-x + 3 = 0$ **(c)** $x^3 - x + 3 = 0$ **(d)** $x^2 - x + 3 = 0$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the discriminant of $2x^2 + x + 3 = 0$ and state the nature of its roots. 2

12. Check whether $x = -4$ is a root of $x^2 - 16 = 0$. 2

13. Find the roots of $(x - 6)(x + 1) = 0$. 2

14. Find the discriminant of $3x^2 - 7x + 2 = 0$ and state the nature of its roots. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by factorisation: $x^2 - 9x + 8 = 0$. 3

16. Find the discriminant of $5x^2 - 10x + 5 = 0$ and hence the nature of its roots. 3

17. The product of two consecutive positive integers is 182. Find the integers. 3

OR

Solve by factorisation: $x^2 + 6x + 8 = 0$. 3

18. Find the discriminant of $2x^2 + 7x + 3 = 0$ and hence the nature of its roots. 3

OR

The product of two consecutive positive integers is 240. Find the integers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Solve by factorisation: $4x^2 - 5x - 6 = 0$. 5

OR

The length of a rectangular plot exceeds its breadth by 8 m and its area is 153 m². Find its dimensions. 5

20. Solve by factorisation: $2x^2 - 3x - 5 = 0$. 5

OR

The length of a rectangular plot exceeds its breadth by 5 m and its area is 204 m². Find its dimensions. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND